

Phase 1: Discovery & Hype

- **P01: Home Page** – High-energy hub with a search bar, "Ending Soon" carousel with live timers, and a platform trust bar.
- **P02: Listing Grid** – Searchable directory with advanced sidebar filters (price, type, condition) and real-time price badges.
- **P03: Auction Detail** – Full item showcase with high-res gallery, seller trust signals, and the "Enter Console" CTA.
- **P10: Pre-Live Waiting Room** – A "hype" page with a giant countdown clock and real-time count of interested bidders.
- **P12: How it Works** – Educational page with the 10x leverage calculator and 4-step auction cycle infographic.
- **P29: Category Directory** – An A-Z sitemap of all niche auction categories for SEO and easy browsing.

Phase 2: The Bidding Engines

- **P04: English Bidding Console** – The "War Room" with dynamic increment buttons, custom bid fields, and a pulsing 60-second timer.
- **P05: Live Leaderboard** – Real-time scrolling feed of anonymized bids and bidder rankings.
- **P06: Bid History Archive** – Transparency log of every bid placed, available after the auction ends.
- **P07: Dutch Auction Interface** – Descending price ticker with a high-speed "BUY NOW" button.
- **P08: Blind Bid Portal** – Private interface for submitting secret, one-time hidden bids.
- **P09: Blind Reveal Page** – Post-deadline page where hidden bids are unmasked and the winner is celebrated.

Phase 3: User Verification & Wallet

- **P13: Auth Page (OTP)** – Mobile + OTP login gateway with Firebase/Twilio integration.
- **P14: KYC Portal** – Identity verification for Aadhaar/PAN via 3rd-party automated APIs (SurePass/Digio).
- **P15: Wallet / Deposit** – Razorpay-powered payment hub to manage balance and "Bidding Power".

- **P16: Transaction Ledger – Personal "Passbook" showing status-coded entries for all deposits and refunds.**
- **P17: Bidder Dashboard – Personal command center to track live "Winning" or "Outbid" status across all items.**
- **P27: User Settings – Profile management with vernacular toggles (English, Hindi, Gujarati) and 2FA settings.**

Phase 4: Seller Management

- **P18: Seller Studio – Dashboard for sellers to track total revenue and manage active/ended listings.**
- **P19: Create Listing Form – Multi-step form to set up English, Dutch, or Blind auctions with handoff locations.**
- **P11: Seller Profile View – Public storefront showing verification badges and historical success rates.**

Phase 5: Closing & Operations

- **P20: Notifications Center – Dedicated inbox for outbid alerts, payment notices, and system updates.**
- **P21: Handoff Room – Private room for winners and sellers to chat and coordinate the final 90% payment.**
- **P22: Handoff Checklist – Safety and inspection guide used before the physical meeting.**
- **P23: Invoice Generator – Utility for downloading official sales memos and receipts.**
- **P24: Dispute Center – Mediation portal to handle claims regarding failed offline handoffs.**
- **P25: Feedback & Reviews – Post-transaction rating system to build community trust.**
- **P26: Admin Panel – Restricted dashboard for KYC approvals, dispute resolution, and global metrics.**
- **P28: Legal & Tax Hub – Repository for T&C, Privacy Policy, and TDS/GST guidelines.**
- **P30: Error & State Pages – Custom branded pages for 404s or withdrawn listings.**

- **P31: Audit Trail (Tech Log) – Internal millisecond-accurate log of every bid for legal proof.**

The footer is split into four quadrants plus a bottom "Socket" bar:

Quadrant	Contents
Marketplace	Live Auctions, Ending Soon, Categories Sitemap, Featured Sellers.
Help & Trust	How it Works (Leverage Math), Safety Handoff Guide, Dispute Center, FAQs.
Legal	Terms of Service, Privacy Policy, IT Act India Compliance, TDS/Tax Information.
Contact	Support Email, Emergency Helpline, Grievance Officer, Office Address.

Core User Journeys

This section outlines the primary paths users take through the platform to complete transactions.

A. The Buyer's Journey (Discovery to Acquisition)

- 1. Discovery:** Browse items via the Home Page (P01), Listing Grid (P02), or Category Directory (P29).
- 2. Verification & Funding:** Complete Auth (P13) and KYC (P14), then deposit the 10% security fee on the Wallet Page (P15).
- 3. Active Bidding:** Enter the Bidding Console (P04) or Dutch Interface (P07) to compete in real-time.
- 4. Handoff & Completion:** Coordinate meeting details in the Handoff Room (P21), follow the Safety Checklist (P22), and finalize the 90% payment.

B. The Seller's Journey (Listing to Revenue)

- 1. Creation:** Use the Create Listing Form (P19) to set auction parameters and the offline meeting point.
- 2. Management:** Monitor performance and incoming bids through the Seller Studio (P18).
- 3. Fulfillment:** Access the Handoff Room (P21) once a winner is declared and the 10% deposit is secured.
- 4. Finalization:** Confirm receipt of the 90% offline balance to close the transaction and generate an Invoice (P23).

C. The Dispute Path (Safety Protocol)

- 1. Filing:** If a handoff fails, either party raises a claim in the Dispute Center (P24).
- 2. Resolution:** Admins review the Audit Trail (P31) and chat logs to issue a verdict.
- 3. Settlement:** Funds are either refunded to the buyer's Wallet (P16) or released to the seller as a penalty fee.

To ensure security and proper user flow, the platform uses Role-Based Access Control (RBAC). Below is the shortened access matrix for Guests, Buyers, Sellers, and Admins.

1. Guest Access (Unauthenticated)

- **Public Discovery:** Can view the Home Page (P01), Listing Grid (P02), and Auction Details (P03).
- **Educational:** Access to How it Works (P12) and the Legal/Tax Hub (P28).
- **Restricted:** Cannot bid, deposit funds, or view seller profiles. Redirected to Auth (P13) for any action.

2. Buyer Access (Verified Account)

- **Discovery & Bidding:** All Guest pages plus the Bidding Consoles (P04, P07, P08) and Waiting Room (P10).
- **Financials:** Full access to Wallet/Deposit (P15) and Transaction Ledger (P16).
- **Management:** Access to the Bidder Dashboard (P17), Notifications (P20), and User Settings (P27).
- **Post-Auction:** Access to the Handoff Room (P21), Checklist (P22), and Reviews (P25) for won items.

3. Seller Access (Verified Seller)

- **Storefront Management:** All Buyer pages plus the Seller Studio (P18) and Create Listing Form (P19).
- **Public Identity:** Can manage and view their own Seller Profile (P11).
- **Fulfillment:** Access to the Handoff Room (P21) and Invoice Generator (P23) for sold items.

4. Admin Access (Internal Control)

- **Global Oversight:** Exclusive access to the Admin Panel (P26) for KYC approvals and revenue tracking.
- **Mediation:** Can enter the Dispute Center (P24) to review evidence and issue verdicts.
- **Technical Proof:** Access to the Audit Trail (P31) to track millisecond-accurate logs for legal disputes.
- **System Integrity:** Can view and manage all Error & State Pages (P30) and the Category Directory (P29).

P01: Home Page

The entry point must be high-energy and show that the marketplace is live and trustworthy.

- **Header & Navigation:**
 - Sticky header for laptop users.
 - Search bar with category dropdown.
 - Live "Activity Badge" in the corner (e.g., "1,240 Bids in the last hour").
- **Hero Section:**
 - Clean, bold typography.
 - Primary CTA: "Start Bidding" (links to P02) and "Start Selling" (links to P18).
- **"Ending Soon" Section (The Hook):**
 - A horizontal carousel of cards.
 - **Live Timer:** Each card has a countdown that turns red when under 5 minutes.
- **Category Grid:**
 - Visual tiles for main categories (Electronics, Art, Vehicles, etc.).
- **Platform Trust Bar:**
 - Icons for "Verified Users," "Secure 10% Escrow," and "Offline Handoff Safety."

Backend/API:

- GET /api/auctions/featured: Returns a list of high-value or trending auctions.
- GET /api/auctions/ending-soon: Returns auctions closing within the next 60 minutes.

P02: Listing Grid Page

This is the "Search & Filter" hub designed for power browsing.

- **Layout:**
 - Left Sidebar: Filters (Price range, Auction Type: English/Dutch/Blind, Condition, Location).
 - Main Content: 4-column grid of Auction Cards.
- **Auction Card Features:**
 - Live Price Badge: Updates via Socket.io when a new bid is placed.
 - "Quick Watch" Button: Adds the item to the user's dashboard without leaving the page.
 - Progress Bar: Visual representation of time remaining (e.g., green for 24h+, orange for <2h, red for <10m).
- **Empty State:**
 - If no results match filters, suggest "Saving this search" to get an alert when an item matches.

Backend/API:

- GET /api/auctions: Supports robust query parameters (e.g., category, minBid, maxBid, type).
- **Pagination:** Use infinite scroll or "Load More" for a smooth laptop experience.

P03: Auction Detail Page (Discovery Mode)

This is the showcase page where a user decides if they want to commit their deposit.

- **Media Gallery:**
 - Large high-res primary image.
 - Thumbnail strip below for multiple angles.
 - Hover-to-zoom (essential for inspecting item condition on a laptop).
- **Bidding Stats Card:**
 - Current Bid, Number of Bidders, and Starting Price.
 - Countdown Timer (Large and synchronized).
- **Seller Credibility Card:**
 - Verified Badge, Rating, and "Success Rate" (Handoffs completed).
 - Link to **P11 (Seller Profile)**.
- **Description & Information Tabs:**
 - Tab 1: Detailed Description.
 - Tab 2: Handoff Location (e.g., "Public location in Surat").
 - Tab 3: Platform Rules (Reminder of the 10% deposit requirement).
- **The "War Room" CTA:**
 - A prominent button: **"Enter Bidding Console"**.
 - *Logic Check:* If user is not logged in $\rightarrow$ Redirect to P13. If not verified $\rightarrow$ Redirect to P14.

Backend/API:

- GET /api/auctions/:id: Fetches all metadata, including images and full seller profile links.

P04: English Bidding Console (The "War Room")

This is a dedicated, high-performance view. On a laptop, this should feel like a trading terminal—fast, responsive, and data-rich.

1. Primary UI Components:

- **The Focus Gallery:** A condensed version of the item images (taking up less space than P03 to prioritize bidding tools).
- **Live Bid Input Zone:**
 - **Smart Buttons:** Three large buttons showing Current Bid + Increment. (e.g., if the bid is ₹1,000 and the increment is ₹100, buttons show: ₹1,100, ₹1,200, ₹1,300).
 - **Custom Bid Field:** A manual entry box with "Max Bid" validation.
- **Real-Time Status Bar:**
 - **The Timer:** A large digital clock. When it hits <60s, it pulses red.
 - **Bidding Power Meter:** A progress bar showing: *"You have used ₹X of your ₹Y bidding limit."*
- **System Alerts:** Small pop-ups within the console for: *"You have been outbid!"* or *"Auction extended by 2 minutes."*

2. Socket.io Event Map (Client to Server):

- `emit('place_bid', { auctionId, amount })`: Sends the bid attempt.
- `on('bid_rejected')`: If the backend logic (10x rule) fails, the UI shakes and shows an error without reloading.
- `on('bid_accepted')`: Updates the local state immediately (Optimistic UI).

P05: Live Leaderboard (The Transparency Feed)

This usually sits as a sidebar or a secondary panel on the laptop screen, providing the "social proof" of the auction.

1. Primary UI Components:

- **The High-Bid Header:** Displays the anonymized username (e.g., S***m_98), their city, and their trust rating.
- **The Scrolling Bid Feed:** A vertical list of every bid placed in the last 10 minutes.
 - *Format:* [Timestamp] - [Username] - [Amount].
 - *Visuals:* New bids "slide in" from the top with a brief highlight effect.
- **Bidder Count:** A small counter showing: *"14 people currently bidding."*
- **Increment Table:** A small reference table showing what the minimum raise is for the current price bracket.

2. Backend & Socket Logic:

- **Broadcast Event:** When any user successfully bids, the server emits broadcast('new_bid_update') to **all** users in that auction room.
- **Redis Integration:** The "Current Highest Bid" is stored in Redis for sub-millisecond retrieval so the leaderboard never lags, even with hundreds of bidders.

P06: Bid History Archive (Transparency Page)

Once the timer hits zero, the "War Room" (P04) closes, and participants are redirected here. This page serves as the official legal record of the event.

1. Primary UI Components:

- **The "Final Hammer" Header:** Displays the winning bid amount, the winner's anonymized ID, and the total number of bids placed.
- **The Full Bidder Ranking:** A table listing every unique participant.
 - *Columns:* Rank, Bidder ID (Anonymized), Highest Bid Submitted, and Timestamp of their last bid.
 - *Logic:* Even if someone bid 50 times, only their **highest** successful bid is shown here to keep the list clean.
- **Auction Timeline:** A linear chart showing how the price rose over time. This helps prove that the "Popcorn Bidding" extensions were triggered fairly.

2. Backend & Data Logic:

- **GET /api/auctions/:id/history:** Fetches the sorted list of highest bids per user from the Bids collection.
- **Data Integrity:** This page is "Read-Only." Once the auction status is ENDED, these records are locked to prevent any after-the-fact manipulation.

P07: Dutch Auction Interface (The "Price Drop" Page)

The Dutch auction is the opposite of the English one. It's a game of "Who clicks first?" This requires a completely different UI.

1. Primary UI Components:

- **The Descending Price Display:** A massive, centrally located price that updates in real-time.
 - *Visuals:* The numbers "tick" down like a stopwatch.
- **The "Price Drop" Schedule:** A sidebar showing exactly when the next drop will happen (e.g., *"Price drops by ₹500 in 12 seconds"*).
- **The "BUY NOW" Button:** Instead of an increment button, there is one giant button.
 - *Status Check:* This button is only active if the user has enough **Bidding Power** (10% of the *current* price).
- **Real-Time Remaining Quantity:** If the seller is selling multiple identical items (common in Dutch auctions), a counter shows how many are left.

2. Technical Mechanics (Socket.io & Redis):

- **Server-Side Ticking:** The server uses a `setInterval` or a Redis-backed timer to broadcast the new price to all clients every X seconds.
- **The "Race Condition" Shield:** Since multiple people might click "BUY" at the exact same millisecond, the backend must use **Redis Locking**. The first request to hit the server wins; all others are immediately served a "Sold Out" message.

P08: Blind Bid Portal (The "Secret" Page)

Used for high-value collectibles or government-style tenders where you don't want bidders to influence each other.

1. Primary UI Components:

- **The Private Input:** A single, secure field to enter a "Max Bid."
- **The "Sealed" Animation:** When the user submits, the UI shows a visual of an envelope being sealed to reinforce the privacy of the bid.
- **Edit Functionality:** Users can update their bid as many times as they want *until* the deadline, but they can never see anyone else's amount.

2. Backend/Security:

- **Encryption:** Bids are stored in the database but are hidden from the "Current Bid" field of the Auction object until the revealDate.
- **Validation:** The system still checks the 10x leverage rule against the secret bid to ensure the bidder is "good for the money."

P09: Blind Reveal Page (The "Unsealing")

In a Blind/Sealed-bid auction, this page is the climax. It transforms a private process into a transparent public record once the timer hits zero.

1. Primary UI Components:

- **The Unveiling Animation:** A one-time visual effect (like a digital seal breaking) that triggers when a participant first visits the page after the reveal date.
- **The Ranked Leaderboard:**
 - **Winner Spotlight:** A large, celebrated card at the top showing the winning bid and the bidder's anonymized ID.
 - **The "Full Spread":** A table showing all submitted bids from highest to lowest.
 - **Logic:** This proves to the 2nd and 3rd place bidders exactly how much they lost by, ensuring they don't feel the system was "rigged."
- **Refund Status Badge:** For all losers, a clear status indicator: *"Refund of ₹X initiated to your wallet/bank."*

2. Backend Logic:

- **Atomic Reveal:** The backend flips a `isRevealed` boolean in the database. Until this is true, the bids collection is restricted from all GET requests except for the Admin.
- **Audit Trail:** A timestamped log showing that the bids were only decrypted/unmasked *after* the submission deadline.

P10: Pre-Live Waiting Room (The "Hype" Page)

For scheduled high-value auctions (e.g., "Starts Friday at 6 PM"), you don't want users staring at a static "Not Found" page. You want them to "check in."

1. Primary UI Components:

- **The Giant Countdown:** A high-precision clock ticking down to the start second.
- **"Register to Bid" Button:**
 - *Logic:* This triggers the 10% deposit flow (P15) early so the user is ready the second the auction starts.
- **Participation Count:** Shows *"450 people are watching this item"* to build social proof and competition.
- **Item Sneak-Peek:** A limited gallery and "Key Specs" summary to keep the user engaged while they wait.
- **Add to Calendar:** Integration for Google/Outlook calendars.

2. Technical Mechanics:

- **Socket.io "Ready" Event:** When the timer hits zero, the server emits a `room_open` event. The client automatically transitions from P10 to P04 (The Bidding Console) without the user needing to refresh.
- **Pre-Auth Check:** The page checks if the user's KYC is complete. If not, it shows a warning: *"You have 15 minutes to complete KYC to participate in this auction."*

P11: Seller Profile View (Public)

This is the page we discussed earlier—the "Storefront" for the seller that builds trust for the 10% deposit.

1. Primary UI Components:

- **Trust Stats:** Success rate (completed handoffs), total items sold, and account age.
- **Verified Badges:** Visual icons for ID, Phone, and Email verification.
- **Current Listings:** A mini-grid of other items this seller is currently auctioning.
- **Review Feed:** Aggregated star ratings and text reviews from previous winners.

2. Backend API:

- `GET /api/users/:id/profile`: A public-facing route that excludes sensitive data like the seller's actual PAN/Aadhaar reference or bank details, showing only the "Aggregated Reputation" data.

P12: How it Works (Educational Hub)

Before users give you their data or money, they need to understand the "Ground Rules." This page reduces support tickets by explaining the 10x leverage math clearly.

1. Primary UI Components:

- **The "Rule of 10" Calculator:** An interactive slider where a user inputs a deposit amount (e.g., ₹5,000) and it visually shows their "Bidding Power" (₹50,000).
- **The Auction Cycle Infographic:** A 4-step visual:
 1. Verify & Deposit.
 2. Bid in Real-Time.
 3. Win & Capture (10%).
 4. Offline Handoff (90%).
- **Video Tutorial:** A short (60s) screen recording of a mock English auction.
- **Safety FAQ:** "What happens if I lose?" (Refund) and "What if the seller is a scammer?" (Dispute Center).

2. UX Focus:

- **Laptop Layout:** Use a "Sticky Sidebar" for quick navigation between different auction types (English vs. Dutch vs. Blind).

P13: Auth Page (Firebase / JWT Entry)

This is the "Gatekeeper." Since we are dealing with financial transactions, we use **Phone + OTP** as the gold standard for Indian marketplaces.

1. Primary UI Components:

- **Mobile Number Input:** With a country code prefix (+91).
- **OTP Verification Field:** A 6-digit input with an "Auto-focus" feature and a "Resend in X seconds" timer.
- **Social Login (Optional):** Google/Apple login for faster onboarding, but phone verification remains mandatory before bidding.
- **Legal Consent:** Checkboxes for "I agree to the T&C and the 10% Deposit Rule."

2. Backend/Security Logic:

- **Integration:** Use Firebase Auth or a service like Twilio/Msg91 for OTP.
- **JWT Storage:** Upon successful OTP, the backend issues a **JSON Web Token** stored in an httpOnly cookie for secure session management.
- **User Document Creation:** If it's a new number, create a User doc in MongoDB with kycStatus: 'NOT_STARTED' and walletBalance: 0.

P14: KYC Portal (Identity Verification)

This is the most critical page for platform trust. You don't want "bot" bidders or fake accounts.

1. Primary UI Components:

- **ID Type Selector:** Dropdown for Aadhaar or PAN.
- **Document Upload Zone:** A drag-and-drop area for front and back images of the ID.
- **Live "Liveness" Check (Optional):** A laptop webcam prompt to take a selfie to match against the ID photo.
- **Verification Status Stepper:** Uploaded $\rightarrow$ Under Review $\rightarrow$ Verified.

2. Backend & API Integration:

- **Third-Party API:** Use a service like **SurePass**, **Digio**, or **HyperVerge** to verify the ID against government records.
- **Security Directive: Never** store the raw ID digits in your own database. Store only the "Verification Reference ID" provided by the service and the user's name as it appears on the ID.
- **Status Update:** Once the API returns "Success," update the user's MongoDB record to kycStatus: 'VERIFIED'.

P15: Wallet / Deposit Page

This page is the bridge between the user's bank account and their participation in an auction. It must feel as secure as a banking app.

1. Primary UI Components:

- **The Power Dial:** A visual gauge showing **Current Balance** vs. **Total Bidding Power** ($\text{\$Balance} \times 10\%$).
- **Quick-Add Buttons:** Pre-set amounts for easy top-ups (e.g., ₹1,000, ₹5,000, ₹10,000).
- **Razorpay Checkout Integration:** A "Add Funds" button that triggers the Razorpay popup.
- **Live Leverage Calculator:** An input where a user types a target bid (e.g., "I want to bid ₹50,000") and the UI tells them exactly how much more they need to deposit (₹5,000 total deposit required).

2. Backend Logic (Razorpay Flow):

- **The "Authorization" Strategy:** Use Razorpay's **Manual Capture** feature. When a user deposits, the payment is "Authorized" but not "Captured" until they actually win an auction.
 - *If they lose:* The authorization is released (voided), and the money returns to the user faster than a standard refund.
- **Atomic Updates:** Use a MongoDB transaction to ensure that `user.walletBalance` and the Transaction record are updated together.

P16: Transaction Ledger (The "Passbook")

In a marketplace with complex deposit/refund rules, users will constantly check their history to see where their money is.

1. Primary UI Components (2026 Fintech Standards):

- **Status Badges:** Color-coded tags for every entry:
 - HELD: 10% deposit locked for an active bid.
 - REFUNDED: Deposit returned after losing an auction.
 - CAPTURED: 10% successfully paid for a won item.
 - TOP-UP: Direct deposit into the wallet.
- **Timeline View:** A vertical list showing the Date, Auction Name, and Amount. Clicking an item expands it to show the **Transaction ID** and a "Download Receipt" link.
- **Net Balance Header:** Large, bold numbers for "Total Deposited" vs. "Currently Locked."

2. Data Structure:

- **GET /api/transactions:** Should return a list sorted by createdAt: -1.
- **Receipt Generator:** A small microservice that takes the Transaction ID and generates a PDF invoice (P23).

P17: Bidder Dashboard (Central Command)

This is the user's personal "Control Tower." If they are bidding on 5 things at once, they manage them here.

1. Primary UI Components:

- **"Live Bids" Section:** Cards for every auction the user is currently in.
 - *Winning:* Green border.
 - *Outbid:* Pulsing red border with a "Re-bid Now" button.
- **Watchlist:** Items the user has marked but hasn't bid on yet.
- **Won Auctions (Pending Handoff):** Items where the user won but the 90% offline payment isn't finished. This card links directly to the **Handoff Room (P21)**.

2. UX Focus:

- **Real-Time Polling:** While not as fast as the War Room (P04), this dashboard should use Socket.io to update the "Winning/Outbid" status the moment someone else bids.

P18: Seller Studio (The Seller Dashboard)

This is the private command center for anyone selling on your platform. It's designed to give a birds-eye view of their revenue and active listings.

1. Primary UI Components:

- **Sales Overview Tiles:**
 - **Live Revenue:** Potential earnings from active auctions.
 - **Pending Handoffs:** Number of items waiting for the 90% offline payment.
 - **Completed Sales:** Total history of successful transactions.
- **Auction Management Tabs:**
 - **Active:** Items currently being bid on.
 - **Drafts:** Listings started but not yet published.
 - **Ended:** Archive of past auctions.
- **The "Action Center":** A list of tasks for the seller (e.g., *"3 auctions ended. Click here to contact winners"* or *"Update KYC to receive payments"*).

2. Backend Logic:

- **GET /api/seller/dashboard:** Aggregates the seller's specific metrics using MongoDB \$facet or multiple count queries.
- **Authorization:** All routes here must verify that the User has isSeller: true permissions in their JWT.

P19: Create Listing Form (Multi-Type Setup)

This is the most complex form in your app. It must adapt dynamically based on the **Auction Type** the seller chooses.

1. Primary UI Components:

- **Step 1: Item Basics:** Title, Category, and Multi-Image Uploader (with drag-and-drop ordering).
- **Step 2: Auction Type Selector:** A choice between **English**, **Dutch**, or **Blind**.
- **Step 3: Dynamic Pricing Logic:**
 - **If English:** Fields for *Starting Price*, *Reserve Price*, and *Bid Increment*.
 - **If Dutch:** Fields for *Starting Price*, *Price Floor*, and *Drop Interval* (e.g., Drop by ₹500 every 1 minute).
 - **If Blind:** Field for *Submission Deadline* and *Reveal Time*.
- **Step 4: Handoff Details:** A text area or map picker to specify the **Offline Meeting Point** (crucial for your model).

2. Form Logic & Validation:

- **The 10% Calculation:** The UI should show the seller: *"If this sells for ₹10,000, we will hold ₹1,000 as a deposit from the buyer."*
- **Validation:** Prevent the seller from setting a "Price Floor" higher than the "Starting Price" in a Dutch auction.

P20: Notifications Center

In a live auction, speed is everything. If a user is outbid, they need to know in milliseconds.

1. Primary UI Components:

- **Priority Filters:** "All", "Bids", "Payments", and "System Alerts."
- **Actionable Items:** Each notification should be a link.
 - *Example:* "You were outbid on **Sony PS5**" $\rightarrow$ Links directly to **P04 (Bidding Console)**.
- **Real-Time Toast Alerts:** Small pop-ups that appear regardless of which page the user is on.

2. Technical Stack:

- **Frontend:** A global React Context or Redux store that listens for Socket.io events.
- **Backend:** Use **Firebase Cloud Messaging (FCM)** for push notifications to mobile devices and **WebSockets** for the laptop browser.
- **Persistence:** Every notification is saved in a Notifications collection in MongoDB so the user can see their history later.

P21: Handoff Room (The Private War Room)

This page is only accessible to the winner and the seller once the 10% deposit has been successfully captured.

1. Primary UI Components:

- **The Contact Exchange:** Displays the verified phone number and name of both parties.
- **Secure Chat Interface:** A dedicated chat thread (Socket.io) to coordinate the time and location of the meeting.
- **Handoff Status Stepper:**
 - Contacted $\rightarrow$ Meeting Scheduled $\rightarrow$ Payment Received $\rightarrow$ Item Received.
- **The "Payment Received" Button (Seller Only):** This is the "kill switch." When the seller clicks this, it confirms they have the 90% in cash/UPI, and the transaction closes.
- **The "Raise Dispute" Button:** Becomes active only after 48 hours if the transaction hasn't been closed.

2. Backend Logic:

- **Privacy Guard:** Ensure User_A cannot see User_B's details until the Auction.status is officially SOLD.
- **Logging:** Every message in this chat is saved. If a dispute occurs later, the Admin uses this log as evidence.

P22: Handoff Checklist (Safety & Inspection)

This is a "Gatekeeper" page that appears as a popup or a mandatory read before the Handoff Room opens. It protects you from liability.

1. Primary UI Components:

- **Inspection Guide:** A dynamic list based on the category (e.g., for Electronics: "Check battery health," "Verify screen pixels").
- **Safety Protocols:** "Meet in a well-lit public place," "Do not go alone," "Verify the 90% payment before handing over the item."
- **"I Understand" Checkbox:** The user must acknowledge that the platform is not responsible for the item's condition once they leave the meeting.

P23: Invoice / Receipt Generator

For high-value auctions (like the ones you're targeting in 2026), users need a paper trail for tax or warranty purposes.

1. Primary UI Components:

- **Breakdown Table:**
 - Hammer Price (100%).
 - Security Deposit Paid to Platform (10%).
 - Balance Paid Offline (90%).
- **Seller & Buyer Details:** Names and anonymized IDs.
- **Download Buttons:** "Save as PDF" or "Print Receipt."

2. Technical Implementation:

- **Library:** Use jspdf or react-pdf on the frontend, or puppeteer on the backend for high-quality generation.
- **Watermarking:** Add a "Verified by [Your Platform Name]" watermark to prevent tampering.

P24: Dispute Center (The Digital Courtroom)

This page is the only thing that protects your platform's 2026 reputation. It manages the 10% deposit if one party claims the other didn't show up or lied about the item.

1. Primary UI Components:

- **The "Claim" Form:** A multi-step flow where the user selects a reason (e.g., "Item not as described," "Seller didn't show up," "Buyer refused to pay 90%").
- **Evidence Uploader:** A drag-and-drop zone for chat screenshots, photos of the item taken at the meeting spot, or even a video of the item's defect.
- **Case Status Tracker:** A horizontal stepper showing: Dispute Raised $\rightarrow$ Admin Assigned $\rightarrow$ Evidence Review $\rightarrow$ Verdict.
- **Mediation Chat:** A specialized chat window where an Admin can talk to both the Buyer and Seller simultaneously to settle the issue.

2. Backend & Verdict Logic:

- **The Refund/Release Trigger:**
 - *Scenario A:* Admin finds the Seller at fault $\rightarrow$ 10% is refunded to the Buyer.
 - *Scenario B:* Admin finds the Buyer at fault $\rightarrow$ 10% is released to the Seller (as a "no-show fee").
- **Audit Trail:** Every action taken by the Admin is logged in **P31 (Audit Trail)** for legal compliance under Indian E-commerce rules.

P25: Feedback & Reviews (Reputation Building)

Trust is the currency of your platform. This page ensures that "Good Sellers" rise to the top and "Bad Actors" are filtered out.

1. Primary UI Components:

- **Star Rating (1–5):** Categorized ratings for "Item Accuracy," "Communication," and "Punctuality."
- **Verified Purchase Badge:** A visual tag on the review to prove this person actually won the auction and paid the deposit.
- **Photo Reviews:** Allows the buyer to upload a "Handover Photo" of the item they just bought.
- **The "Report" Flag:** A small button for the seller to report a "Troll Review."

2. Data Logic:

- **Aggregate Scoring:** When a review is submitted, a backend hook recalculates the `averageRating` for the user in the `Users` collection.
- **Cooldown Period:** Reviews can only be submitted *after* the "Payment Received" button is clicked in P21, preventing fake reviews.

P26: Admin Panel (The Master Control)

This is the most powerful page in your MERN app. It's a restricted-access dashboard for you (the owner) to manage the entire ecosystem.

1. Primary UI Components:

- **The "Global Pulse":** Real-time stats showing Total Live Auctions, Total HELD Deposits (in INR), and Pending KYC requests.
- **User Management Table:** Search for any user, view their KYC docs, and manually "Verify" or "Ban" them.
- **Dispute Queue:** A list of all open disputes sorted by urgency/time-passed.
- **Revenue Tracker:** Shows the 2% platform fee you've collected from every successful 10% capture.

2. Security Requirement:

- **RBAC (Role-Based Access Control):** This page must be protected by a middleware that checks `user.role === 'ADMIN'`. Use multi-factor authentication (MFA) for this specific page.

P27: User Settings (The Preferences Hub)

Since you are targeting a diverse user base across India (Surat, Vadodara, and beyond), this page is about personalization and accessibility.

1. Primary UI Components:

- **Profile Management:** Update display name, profile picture, and verified email.
- **Vernacular Toggle (2026 Special):** A dropdown to switch the entire UI between **English, Hindi, and Gujarati**.
- **Security Settings:** Change password and manage **Two-Factor Authentication (2FA)** for high-value bidding.
- **Notification Toggles:** Granular controls for SMS, Email, and Browser Push (e.g., "Only notify me if outbid in the last 5 minutes").

2. Backend Logic:

- **PATCH /api/users/me:** Updates specific fields in the MongoDB user document.
- **Language Persistence:** Store the preferredLanguage in the database and the user's local storage so the UI remains in their chosen language across devices.

P28: Legal & Tax Hub (Compliance Layer)

For an auction platform in India, you must comply with the **IT Act** and **TDS (Tax Deducted at Source)** rules for high-value items.

1. Primary UI Components:

- **Terms & Conditions:** A searchable document detailing the 10% deposit and 90% offline handover rules.
- **Tax Calculator:** A small tool showing the user any applicable GST or TDS (for transactions over ₹50,000) that they need to be aware of during the offline handoff.
- **Privacy Policy:** Clear explanation of how KYC data is handled (referencing your "Zero-Storage" policy for raw ID digits).
- **Grievance Officer Info:** Legally mandated contact details for dispute escalation.

P29: Category Directory (The Sitemap)

While search is primary, a directory helps with SEO and allows users to discover niche high-value markets.

1. Primary UI Components:

- **A-Z Index:** A clean, multi-column list of all categories (e.g., Antiques, Business Machinery, Designer Tech).
- **Popular Categories:** Visual cards for the top-performing sectors.
- **"Request a Category":** A small form for users to suggest new types of items they want to sell.

P30: Error & State Pages (UX Glue)

Don't let the user hit a "dead end." These pages keep the flow moving even when things go wrong.

1. Primary UI Components:

- **404 Not Found:** A branded page with a "Go Back to Auctions" button.
- **Auction Withdrawn:** If a seller cancels an auction, this page explains why and confirms that any held deposits have been released.
- **Maintenance Mode:** A professional "We'll be back at [Time]" screen for system updates.

P31: Audit Trail (The Technical Log)

This is an internal-only page (accessible only to Admins) used for legal proof if a bidder claims the system was rigged.

1. Primary UI Components:

- **The "Pulse" Log:** A millisecond-accurate table of every event: Bid Placed $\rightarrow$ Socket Confirmed $\rightarrow$ Database Written $\rightarrow$ Winner Triggered.
- **IP & Device Tracking:** Shows the source of bids to identify and block "Shill Bidding" (sellers bidding on their own items from fake accounts).
- **Data Export:** Allows the admin to export a "Legal Proof PDF" for court or police use if a transaction turns into a criminal fraud case.

